SIMATS ENGINEERING

ASSIGNMENT - 5

Sub Code: CSA10

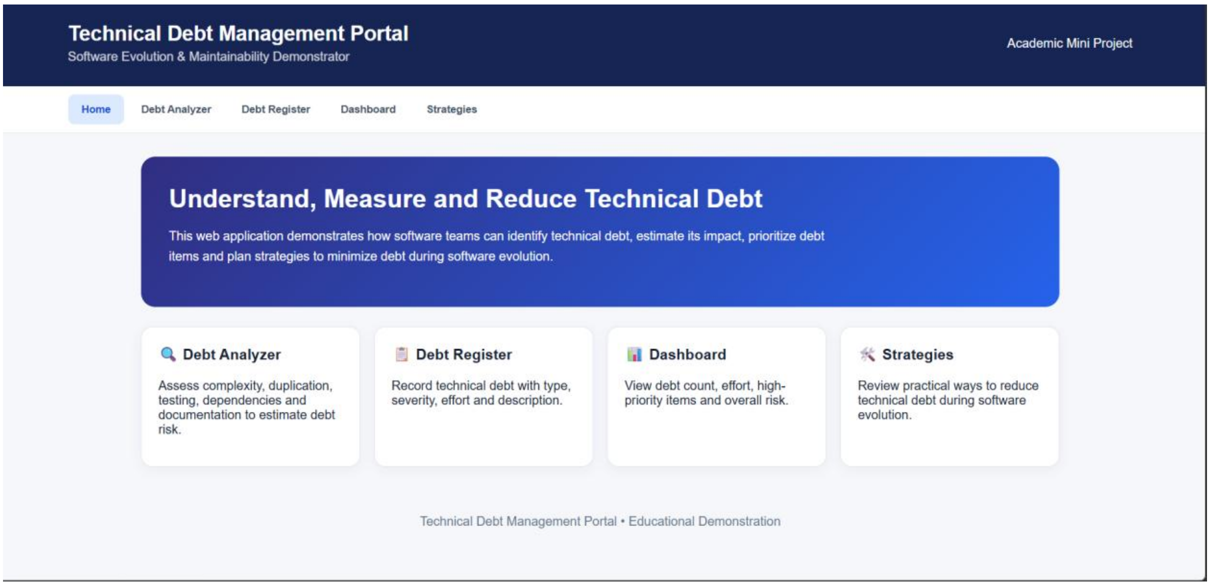
Subject Name: Software Engineering

1. Explain technical debt. What are the major causes of technical debt, and how can organizations minimize it during software evolution?

INTRODUCTION:

Software systems continuously evolve to meet changing business requirements, customer expectations, and technological needs. As software grows, developers sometimes make quick or temporary implementation decisions instead of choosing the best long-term solution. These shortcuts may help deliver software faster, but they can create additional work and maintenance costs in the future. This accumulated cost is known as **technical debt**.

Technical debt is an important concept in software engineering because unmanaged technical debt can make software difficult to maintain, modify, test, and scale. Therefore, organizations need to identify, monitor, and reduce technical debt throughout the software development life cycle



Technical Debt Management Portal
Software Evolution & Maintainability Demonstrator

Academic Mini Project

Home | **Debt Analyzer** | Debt Register | Dashboard | Strategies

Technical Debt Analyzer

Project Assessment

Code complexity (1–10)
6

Code duplication (%)
20

Test coverage (%)
65

Outdated dependencies
4

Documentation quality
Good

Analyze Technical Debt

Assessment Result

Estimated Debt Risk Score
68

MEDIUM RISK

Interpretation: Schedule refactoring and testing improvements before debt grows.

Home | Debt Analyzer | **Debt Register** | Dashboard | Strategies

Technical Debt Register

Debt item
e.g. Duplicate payment logic

Severity
Low

Type
Code

Estimated effort (hours)
8

Description
Describe the problem and future impact

Add Debt Item

Current Debt Register

Debt Item	Type	Severity	Effort	Description	Action
Duplicate validation logic	Code	Medium	10h	Same validation exists in several modules.	Remove
Low automated test coverage	Testing	High	20h	Important modules lack regression tests.	Remove

Home | Debt Analyzer | Debt Register | **Dashboard** | Strategies

Technical Debt Dashboard

Total Items	Estimated Effort	High/Critical	Overall Risk
3	38h	1	Medium

Debt Distribution

Code — 2 item(s)

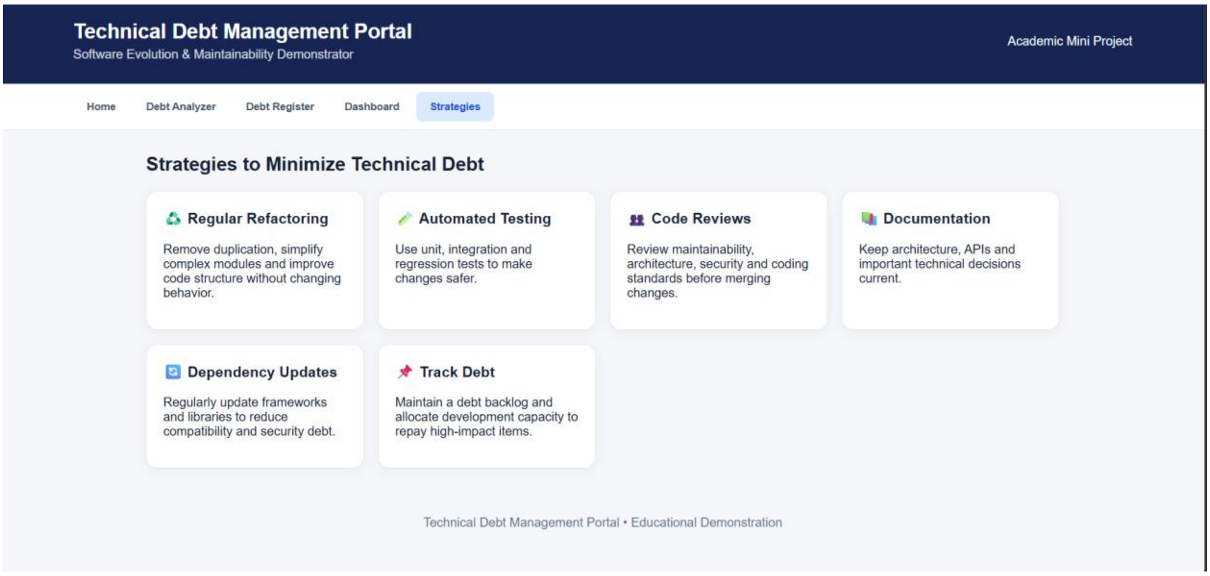
Testing — 1 item(s)

Recommended Action

PRIORITY ACTION

Address High and Critical items first, then schedule refactoring, automated testing, dependency updates and documentation work.

Technical Debt Management Portal • Educational Demonstration



RESULT:

The Technical Debt Management web application was successfully developed to identify, record, analyze, and prioritize technical debt during software evolution. The Debt Register was populated with different types of technical debt, including outdated third-party libraries, low automated test coverage, large monolithic modules, duplicate validation logic, and missing API documentation. The results showed that Critical and High severity items require immediate attention because they can negatively affect software security, maintainability, reliability, and future development speed. The dashboard successfully calculated the total number of debt items, estimated effort required for resolution, and overall technical debt risk. The application also provided recommendations to prioritize high-impact issues and use practices such as refactoring, automated testing, code reviews, documentation, and dependency updates. Overall, the system demonstrates that continuously monitoring and managing technical debt can help organizations maintain high-quality, secure, reliable, and maintainable software throughout its evolution.